Debre Berhan University

College of Computing

Department
of
Information Technology

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*Target group**: 3rd year Information Technology***

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Chapter 5:
Java Frameworks

Overview of Java frameworks

- **Java has a lot of frameworks such as:**
  - **Web Frameworks**
  - **Persistence Frameworks**
  - **Testing Frameworks**
  - **Microservices Frameworks**
  - **UI Frameworks**
  - **Networking & Messaging Frameworks**

Overview of Java frameworks

- **1. Web Frameworks**
  - A web framework is a software framework specifically designed to help developers build web applications and services.
  - It provides pre-built tools, libraries, and structures that streamline the development process, making it faster and easier to create web applications.
- **a. Spring Framework / Spring Boot**
  - Spring Framework and Spring Boot are both Java-based frameworks used for developing applications, but they differ in their scope and purpose.
  - Spring Framework is a comprehensive framework for building enterprise applications, while Spring Boot is a more streamlined framework designed for faster and easier development of Spring-based applications, particularly web applications and microservices.

Spring Framework / Spring Boot…

- They use for Full-stack development, web apps, REST APIs, microservices
- They have the following features :
  - Dependency Injection (IoC),
  - Spring MVC for web apps,
  - Spring Boot simplifies configuration and deployment,
  - Built-in support for REST, data access, and security.

Spring Boot…

- **Advantages**
- Spring Boot offers the following advantages to its developers:
  - Easy to understand and develop spring applications
  - Increases productivity
  - Reduces the development time
- **Goals**
- Spring Boot is designed with the following goals:
  - To avoid complex XML configuration in Spring
  - To develop a production ready Spring applications in an easier way
  - To reduce the development time and run the application independently
  - Offer an easier way of getting started with the application

Spring Boot…

- **Why Spring Boot?**
- You can choose Spring Boot because of the features and benefits it offers as given here:
  - It provides a flexible way to configure Java Beans, XML configurations, and Database Transactions.
  - It provides a powerful batch processing and manages REST endpoints.
  - In Spring Boot, everything is auto configured; no manual configurations are needed.
  - It offers annotation-based spring application
  - Eases dependency management
  - It includes Embedded Servlet Container

Spring Boot…

- **How does it work?**
- Spring Boot automatically configures your application based on the dependencies you have added to the project by using **@EnableAutoConfiguration** annotation.
  - For example, if MySQL database is on your classpath, but you have not configured any database connection, then Spring Boot auto-configures an in-memory database.
- The entry point of the spring boot application is the class contains **@SpringBootApplication** annotation and the main method.
- Spring Boot automatically scans all the components included in the project by using **@ComponentScan** annotation.

Spring Boot…

- Observe the following code for a better understanding:
  - import org.springframework.boot.SpringApplication;
  - Import org.springframework.boot.autoconfigure.EnableAutoConfiguration;
  - @EnableAutoConfiguration
  - public class DemoApplication {
  - public static void main(String[] args) {
    - SpringApplication.run(DemoApplication.class, args);
    - }
  - }

Web Frameworks…

- **b. Jakarta EE (formerly Java EE)**
  - Jakarta EE, formerly Java Platform, Enterprise Edition (Java EE) and Java 2 Platform, Enterprise Edition (J2EE), is a set of specifications, extending Java SE with specifications for enterprise features such as distributed computing and web services.
  - **It has features like:**
    - Servlets, JSP, JSF, JPA, EJB, JAX-RS
    - Built-in support for REST, security, and messaging

Web Frameworks…

- **c. JSF (JavaServer Faces)**
  - JavaServer Faces (JSF) is a Java-based web application framework that simplifies the development of user interfaces for Java web applications.
  - It's a component-based framework, meaning developers can create reusable UI components, manage their state, and handle events.
  - JSF is designed to make it easier to build, maintain, and test web applications.
  - **It has key features like:** Managed beans, UI components, Facelets

Persistence Frameworks

- A persistence framework is a software framework that manages the process of storing and retrieving data, typically from a database.
- It acts as an intermediary between an application's objects and the database, handling tasks like mapping objects to database tables and managing database interactions.
- **a. Hibernate**
  - It is a framework for persisting or saving Java objects in a database.
  - It's a very popular framework used by a lot of enterprise Java projects.
  - It uses ORM (Object Relational Mapping)
  - **Features that are included in hibernate includes:**
    - Maps Java classes to database tables
    - Handles transactions and caching
    - Reduces boilerplate SQL

Persistence Frameworks …

- **b. JPA (Java Persistence API)**
  - Java Persistence API (JPA) is a specification that manages relational data in Java applications.
  - It provides a standard way for Java objects to interact with databases, simplifying data persistence and retrieval.
  - JPA is not a framework or a library, but rather a set of interfaces and rules that define how persistence should be handled.
  - **It has features like:** Annotations for mapping, query language (JPQL)

Testing Frameworks

- A testing framework provides a structured and standardized approach to test development and execution, ensuring consistency and efficiency.
- It acts as a guide, including tools and best practices, to help testers create and manage test cases effectively.
- This framework helps automate testing, reduce errors, and improve the precision of tests.
- **a. JUnit**
  - JUnit is a popular Java testing framework primarily used for unit testing, allowing developers to write and execute automated tests.
  - It helps ensure that individual units or components of code function as intended, contributing to reliable and bug-free software.
  - **It has features like:** Annotations, assertions, lifecycle methods

Testing Frameworks …

- **b. TestNG**
  - TestNG is a popular open-source testing framework for Java, similar to JUnit and NUnit, but with enhanced features and capabilities.
  - It's designed to make it easier to write, organize, and run automated tests, particularly for web applications when used with Selenium.
  - **It has features of:** Data-driven testing, parallel execution, flexible configuration

Microservices Frameworks

- A microservices framework is a collection of tools, libraries, and conventions that simplify the development, deployment, and management of microservices-based applications.
- It helps developers break down large, monolithic applications into smaller, independent services that communicate with each other.

Microservices Frameworks …

- **a. Spring Cloud**
  – Spring Cloud is a suite of tools and frameworks within the Spring ecosystem that simplifies the development of cloud-native microservices.
  – It provides pre-built solutions for common microservices architectural concerns, building upon Spring Boot and often leveraging Netflix OSS components.
  – **It has features like:** Service discovery, config server, load balancing, circuit breaker

Microservices Frameworks …

- **b. Micronaut**
  - Micronaut is a microservices framework, designed for building modular, easily testable applications, especially microservices and serverless functions.
  - **It has the following key features:**
    - Fast startup,
    - low memory footprint,
    - Dependency Injection (DI),
    - Ahead-of-Time (AOT) compilation

Microservices Frameworks …

- **c. Quarkus**
  - Quarkus is a full-stack, Kubernetes-native Java framework made for Java virtual machines (JVMs) and native compilation, optimizing Java specifically for containers and enabling it to become an effective platform for serverless, cloud, and Kubernetes environments.
  - **It has the following key features:**
    - GraalVM support,
    - developer productivity,
    - reactive programming

UI Frameworks

- UI (User Interface) frameworks are collections of pre-built tools, libraries, and templates used to create and manage the visual and interactive elements of a software application.
- They streamline the development process by providing ready-made components like buttons, forms, and menus, allowing developers to focus on functionality rather than building these elements from scratch.

UI Frameworks …

- **a. JavaFX**
  - JavaFX is a Java library and a GUI toolkit designed to develop and facilitate Rich Internet applications, web applications, and desktop applications.
  - **It has the following key Features:**
    - Scene Graph,
    - CSS styling,
    - FXML

UI Frameworks…

- **b. Vaadin**
  - Vaadin Framework is a UI framework with a server-side programming model, in which all the UI logic and its state live on a server, and a web browser executes only the code of UI widgets.
  - In fact, it is a thin client technology, where a browser reflects only what a server commands and all events are sent to a server.
  - **It has features like:**
    - Drag-and-drop UI builder,
    - component-based

Networking & Messaging Frameworks …

- They are tools/libraries that simplify the development of distributed systems by handling communication between devices, servers, or applications over a network.
- They help manage connections, data transmission, and asynchronous communication, often in high-performance or large-scale environments.
- Networking frameworks help build applications that communicate over protocols like TCP/IP, UDP, or HTTP.
- Messaging frameworks enable **asynchronous communication** between different parts of a system (e.g., microservices) via **message queues or topics**.

Networking & Messaging Frameworks …

- **a. Netty**
  - Netty is **an asynchronous event-driven network application framework** for rapid development of maintainable high performance protocol servers & clients.
  - It is a non-blocking framework. This leads to high throughput compared to blocking IO.
  - **It's key features includes:**
    - TCP/UDP client-server apps,
    - efficient and scalable

Networking & Messaging Frameworks …

- **b. Apache Kafka (Java client)**
  - Apache Kafka is a distributed streaming platform.
  - It facilitates publishing, subscribing to, processing, and storing streams of records in a fault-tolerant and durable manner.
  - The Java client for Kafka allows Java applications to interact with a Kafka broker, enabling the production of messages to topics, consumption of messages from topics, and other administrative tasks.
  - **It has key features like:** High throughput, distributed architecture

Question?